x	$P(x)$	$(x-\mu)$	$(x-\mu)^2$
4	0.25	-1.25	1.5625
5	0.35	-0.25	0.0625
6	0.30	0.75	0.5625
7	0.10	1.75	3.0625

Mean $\mu = \sum_x P(x) \cdot x$

$$\mu = \sum_x P(x) \cdot x$$

$$= (4 \times 0.25) + 5(0.35) + 6(0.30) + 7(0.10)$$

$$= 1.0 + 1.75 + 1.8 + 0.7$$

$$\mu = 5.25$$

Variance

$$\sigma^2 = (0.25 \times 1.5625) + (0.35 \times 0.0625) + (0.30 \times 0.0.5625) + (0.10 \times 3.0625)$$

$$\sigma^2 = 0.390625 + 0.021875 + 0.1687 + 0.30625$$

$$\sigma^2 = 0.8875$$

$$\sigma = \sqrt{0.8875}$$

$$\sigma = 0.9421$$

Sampling distribution of mean

$n = 25$

So

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$$\sigma_{\bar{x}} = \frac{0.9421}{\sqrt{25}} = 0.1884$$

PUACP

CS CamScanner

- Convert to z-score

To find $P(\bar{X} < 5.50)$ to a z-score using formula:-

$$z = \frac{X - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

$$= \frac{5.50 - 5.25}{0.1884} = \frac{0.25}{0.1884}$$

$$z = 1.327$$

- Find probability

$z = 1.327$

$P(z < 1.327) = 0.9077$

Conclution:-

The mean of 25 items will be less than 5.50 is approximately 0.908 or 90.8%.

PUACP

CS CamScanner

(ii) Given:-

$n = 30$ (Sample size)

$\mu = 12$ (population area)

$\sigma^2 = 10$ (population variance)

$N = 300$ (population size)

(a) With replacement

(b) Without replacement

(a) With replacement

- The sample mean ($\mu_{\bar{x}}$) is same as the population mean (μ)

$$\boxed{\mu_{\bar{x}} = \mu = 12}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{10}}{\sqrt{30}}$$

$$\sigma_{\bar{x}} = \sqrt{\frac{1}{3}}$$

$$\boxed{\sigma_{\bar{x}} = 0.577}$$

(b) Without replacement

Here we ajust for finite population using finite population correction

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

$\sigma = \sqrt{10}, n = 30, N = 300$

$$\sigma_x = \sqrt{\frac{10}{30}} \times \sqrt{\frac{300-30}{300-1}}$$

$$= \sqrt{\frac{1}{3}} \times \sqrt{\frac{270}{299}} = 0.577 \times 0.948$$

$$\boxed{\sigma_x = 0.547}$$

Sample mean(μ_x) remain some as the population mean(μ)

$$\boxed{\mu_x = \mu = 12}$$

Confidence interval: $CI = \hat{p} \pm Z \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

(iii) Given:-

$n = 100$

$\hat{p} = \frac{\text{Number of success}}{\text{Total sample size}} = \frac{15}{100} = 0.15$

$Z = 1.96$ (critical value for 95% confidence level)

$$S.E = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$= \sqrt{\frac{(0.15)(1-0.15)}{100}} = \sqrt{\frac{(0.15)(0.85)}{100}}$$

$$= \sqrt{\frac{0.1275}{100}} = \sqrt{0.001275}$$

$$\boxed{S.E = 0.0357}$$

BUACR
CamScanner

Calculate Confidence Interval

$C.I = \hat{p} \pm Z \cdot S.E$

$C.I = 0.15 \pm 1.96(0.0357)$

$C.I = 0.15 + 1.96(0.0357)$	$C.I = 0.15 - 1.96(0.0357)$
$C.I = 0.15 + 0.069972$	$= 0.15 - 0.069972$
$C.I = 0.2199$	$= 0.0801$
$\boxed{C.I = 0.22}$	$\boxed{C.I = 0.08}$

(iv) Given:-

$H_0 : \mu = 18$ (Null hypothesis)

$H_a : \mu < 18$ (Alternative)

$\alpha = 0.01$ $\quad \bar{x} = 15$

$n = 25$ $\quad s = 12$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{15 - 18}{12/\sqrt{25}} = \frac{-3}{12/5} = \frac{-3}{2.4}$$

$\boxed{t = -1.25}$

t critical $_{(0.01, 24)} = -2.429$

PUACP

CamScanner

$t = -1.25$

Critical value $= -2.492$

Since $t = 1.25$ is not less than $t_{critical} = -2.492$ we fail to reject null hypothesis.

Conclusion:-

At 1% level of significance, there is insufficient evidence to conclude that $\mu < 18$.

v) Given:-

Null hypothesis $H_0 : \sigma^2 \leq 7.7953$

Alternative $H_1 : \sigma^2 > 7.7953$

Test statistics for variance:-

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

$n = 10$

$s^2 = 193.0711$

$\sigma_0^2 = 7.7953$

$$\chi^2 = \frac{(10-1)(193.0711)}{7.7953}$$

$\chi^2 = 222.91$

CamScanner

$$\chi^2_{(0.05)(9)} = 16.92$$

- If $\chi^2 \leq \chi^2_{crit}$ fail to reject H_0
- if $\chi^2 > \chi^2_{crit}$, reject H_0

$\chi^2 = 222.91$ and $\chi^2_{crit} = 16.92$

Since $\chi^2 > \chi^2_{crit}$, we reject H_0

Thus data support the claim $\sigma^2 > 7.7953$.

PUACP

CamScanner